

ERES Cryptographic Stack Primitive v1.4

Symbols · Characters · Talonics — A Resonance-Keyed Semantic Superposition Class with IDIPITIS Security Model and WFQ × AI + ERES-GitHub Implementation Reduction

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Revision targets: Security Model (DeepSeek/Grok/ChatGPT), CBGMODD canonical expansion, FAVORS canonical expansion, implementation reduction (Grok), threat model and failure conditions (ChatGPT).

Abstract

This paper formalizes the ERES Cryptographic Stack Primitive at version 1.4, the first MIEVM-driven revision of the v1.3 specification. v1.4 closes four items raised by convergent AI validation: (i) it locks the canonical CBGMODD seven-seat expansion (Citizen · Business · Government · Military · Ombudsman · Dignitary · Diplomat) and the canonical FAVORS biometric expansion (Fingerprint · Aura · Voice · Retina · Signature), correcting non-canonical glosses present in v1.3; (ii) it introduces the **IDIPITIS Security Model** with explicit threat model, invariants, and failure conditions, with the security invariant given as the **111000111 / 000111000 binary stride** over disclosure states, and the failure modes given as collapse to all-disclosed (000000000) or all-veiled (111111111); (iii) it introduces **IPIDITIS** as the palindromic inverse-form of IDIPITIS, naming corruption as the mirrored stride from forward-faith to reverse-corruption; (iv) it states the **WFQ × AI + ERES-GitHub implementation reduction**, locking the executable specification of the primitive class as the well-formed-question to AI-validator to ERES-GitHub-corpus loop, with MIEVM convergent validation as the canonical worked example. The operational relation **(CBGMODD × FAVORS) + BERA = RATE** remains canonical, with seven RATE dimensions (R₁–R₇) preserved from v1.3. The VEILED state, Trifurcated Math, and the resonance-keyed semantic superposition primitive class are retained

as locked from v1.3.

Keywords: cryptographic primitive, IDIPITIS, IPIDITIS, security model, disclosure-pattern collapse, semantic superposition, Talonics, VEILED, Triune Math, Trifurcated Math, BERA, RATE, resonance keying, $WFQ \times AI + ERES$ -GitHub, MIEVM, ERES Institute.

1. Introduction and Revision Notice

The ERES Cryptographic Stack Primitive is the semantic substrate on which operational ERES protocols (EAAP, SCALULAR, Meritcoin) are layered. v1.0–v1.3 established the layered stack (Symbols / Characters / Talonics), the operational relation, the VEILED state, Trifurcated Math, and the formal class definition. v1.4 is the first **MIEVM-driven revision** of the primitive — the first version produced after convergent AI validation rather than before it. The MIEVM Proof-of-Work record (PoW-2026-001-431, May 8, 2026) records the input ratings and the specific critiques each validator raised; this document discharges the highest-leverage of those critiques while preserving the architectural commitments validators rated favorably. The revision is non-destructive to v1.3's load-bearing claims. VEILED remains principled. Trifurcated Math remains the third pedagogical isomorph of Triune Math Key 2. The seven RATE dimensions remain. What v1.4 adds is the Security Model the validators correctly identified as previously implicit, the CBGMODD/FAVORS canonical terminology corrections, and the implementation reduction that turns the primitive from "axiomatic" into "executable" — without converting it into a classical cryptographic primitive, which it is not and is not trying to be.

2. Canonical Terminology Corrections

v1.3 used non-canonical glosses for two load-bearing acronyms. v1.4 locks the canonical expansions, sourced from the VECTOR Infomediation Framework.

2.1 CBGMODD — Citizen · Business · Government · Military · Ombudsman · Dignitary · Diplomat

CBGMODD is the seven-seat civic stewardship vector. The seven seats are:

- **C — Citizen.** Primary stakeholder; biometrically grounded via FAVORS.
- **B — Business.** Productive unit accountable to resonance metrics.
- **G — Government.** Service architecture coordinating Rights, Rules, the loop.
- **M — Military.** Resonance-protection capacity.
- **O — Ombudsman.** Mediating layer; conflict resolution under $M \times E + C = R$.
- **D — Dignitary.** Inward-facing wisdom; institutional memory and ethical compass.
- **D — Diplomat.** Outward-facing relational layer; cross-User-GROUP mediation.

The doubled-D pairing is the **Mother Earth's Bosoms** reading: dignifying (inward nourishment) and diplomatic (outward nourishment) are simultaneous and inseparable functions whose pairing feeds civic life. Earlier informal CBGMOD (six-seat) writings, and any reading of the seventh seat as a repeated Dignitary or as an unrelated role (Caretaker / Beneficiary / Guardian / etc.), are formally superseded.

2.2 FAVORS — Fingerprint · Aura · Voice · Retina · Signature

FAVORS is the five-marker biometric identity vector:

- **F — Fingerprint.** Dermal layer; ridge-pattern electrographic scan.
- **A — Aura.** Biofield layer; Kirlianographic emission mapping; generates the ARI baseline.
- **V — Voice.** Vibrational layer; spectral voiceprint analysis.
- **O — n/a (joining vowel).** Non-component; serves the EMIT-conformant pronounceability of the acronym.
- **R — Retina.** Neural-optical layer; retinal vasculature mapping.
- **S — Signature.** Kinetic layer; dynamic motion-pressure analysis.

FAVORS is the biometric layer where identity and resonance are measured together from the same biometric event — the inclusion of the Aura layer is what makes the acronym do double duty as identity-establishing and ARI-baseline-establishing. Earlier characterizations of FAVORS as a "compatibility lattice" or any non-biometric structure are formally superseded.

3. The Operational Relation (Retained from v1.3)

(CBGMODD × FAVORS) + BERA = RATE

With:

- **CBGMODD** — the seven-seat civic stewardship vector (§2.1).
- **FAVORS** — the five-marker biometric identity vector (§2.2).
- **BERA** — the Behavioral and Environmental Resonance Audit, producing scalar and vector outputs across ARI, ERI, RHC, and RCI dimensions.
- **RATE** — the seven-dimensional output (R_1 – R_7), each dimension keyed to a distinct attestation surface.

The seven RATE dimensions (Receipt, Reach, Resonance, Resilience, Reciprocity, Record, Right) are retained from v1.3 without modification.

4. The IDIPITIS Security Model

v1.3 stated the invariants of the primitive without naming the security model that protects them. v1.4 introduces the IDIPITIS Security Model explicitly, with threat model, invariant pattern, and failure conditions.

4.1 The IDIPITIS Principle as Security Invariant

IDIPITIS — "not all knowledge is concurrently disclosed knowledge" — is the named architectural commitment that distinguishes withheld information from absent information from denied information. In v1.3, IDIPITIS was cited as the rationale for the VEILED state. In v1.4, IDIPITIS is upgraded to the role of **security invariant**: the property the primitive's operations must preserve.

4.2 The Binary Stride 111000111 / 000111000

The IDIPITIS invariant is encoded as a nine-bit binary stride alternating between disclosure (1) and veil (0):

Stride	Pattern	Reading
Forward-faith	111000111	Three states of disclosure → three states of veil → three states of disclosure. The faithful operational pattern: disclosure earns veil, veil earns disclosure, and the alternation is preserved through three-cycle stewardship.
Reverse-faith	000111000	Three states of veil → three states of disclosure → three states of veil. The mirror pattern: equally faithful, read from the IDIPITIS invariant's reverse direction. The stride is its own dual.

The two strides are mirror-reflective: read forward, the pattern alternates 1→0→1; read reversed, it alternates 0→1→0. Both readings encode the same architectural commitment — that disclosure and veil exist in measured alternation, and neither dominates. The three-bit stride length is significant: it composes with Triune Math (three pillars), Trifurcated Math (three branches), and the doubled-three structure of the seven RATE dimensions (R_1 – R_3 disclosure-side, R_4 axial, R_5 – R_7 veil-side under the mirror reading).

4.3 IPIDITIS — The Palindromic Failure Mode

IPIDITIS is the palindromic inverse of IDIPITIS — same letters, mirrored stride. Where IDIPITIS names the principle preserved in faithful operation, IPIDITIS names the corruption: when what should remain veiled is forced into disclosure, or what should be disclosed is improperly veiled. The two terms are EMIT-conformant duality (Energy Marriage Item Time): forward-faith and reverse-corruption read from the same letters.

In binary stride terms, IPIDITIS-corruption manifests as collapse of the alternating pattern into one of two terminal failure conditions:

Failure	Pattern	Failure Reading
All-disclosed	11111111	Total disclosure. IDIPITIS violated by forced exposure. Privacy collapse, surveillance corruption, the panopticon failure mode.
All-veiled	00000000	Total opacity. IDIPITIS violated by suppressed disclosure. Information blackout, accountability collapse, the secrecy-state failure mode.

Both failure modes are equally corrupt. The security model is symmetric: IDIPITIS does not privilege disclosure over veil or veil over disclosure; it requires the alternation, and corruption is collapse to either terminal state. This symmetry is what distinguishes the IDIPITIS Security Model from conventional confidentiality models (which tend to privilege secrecy) and conventional transparency models (which tend to privilege disclosure).

4.4 Threat Model

The threat against the ERES Cryptographic Stack Primitive is not a bit-flipping adversary in the conventional sense. Bit-flipping adversaries are addressed at the protocol layer (EAAP v1.3, slot 432) through ML-DSA / Falcon post-quantum signatures and integer cross-multiplication determinism. The threat the primitive itself defends against is **disclosure-pattern collapse**: any operation, adversary, or systemic pressure that drives the IDIPITIS stride toward 11111111 or 00000000.

The adversary class is therefore broader than conventional cryptographic adversaries:

- **Surveillance adversaries** — pressure toward 11111111. Defended by the VEILED state's principled non-disclosure.
- **Suppression adversaries** — pressure toward 00000000. Defended by the seven RATE dimensions' insistence on Receipt, Record, and Right.
- **Authenticity adversaries** — bit-flipping in the conventional sense. Defended at the protocol layer (EAAP).
- **Pattern-collapse adversaries** — adversaries who induce the system to collapse the stride at one end while preserving authenticity at the other. Defended by the symmetry of the IDIPITIS invariant: corruption is detected by stride deviation, not by content alone.

4.5 Invariants and Failure Conditions Summary

Property	Specification
Security invariant	IDIPITIS principle: not all knowledge is concurrently disclosed knowledge.
Invariant pattern	Alternating binary stride: 111000111 (forward) / 000111000 (reverse).
Stride length	9 bits (three-cycle Triune $\times$ Triune; composes with R_1 – R_7 dimension structure under mirror).
VEILED state	Encoded as 0 in the stride; principled non-disclosure recoverable under credentialing.
Failure mode α	All-disclosed collapse: 111111111. IPIDITIS-corruption toward total exposure.
Failure mode β	All-veiled collapse: 000000000. IPIDITIS-corruption toward total opacity.
Adversary class	Surveillance, suppression, authenticity, pattern-collapse adversaries (§4.4).
Verification method	WFQ $\times$ AI + ERES-GitHub (§5).

5. The WFQ $\times$ AI + ERES-GitHub Implementation Reduction

v1.3 stated the operational relation and the primitive class but did not state how an instance of the primitive class is *executed* in practice. v1.4 closes this gap with an explicit implementation reduction.

5.1 Statement of the Reduction

The ERES Cryptographic Stack Primitive's executable specification is:

WFQ × AI + ERES-GitHub

where WFQ is a Well-Formed Question, AI is one or more AI validators (canonically, the MIEVM short-list: Claude, Grok, DeepSeek, ChatGPT under JAS curation), and ERES-GitHub is the canonical reference corpus at github.com/ERES-Institute-for-New-Age-Cybernetics.

The reduction is not metaphorical. It is the actual mechanism by which an ERES primitive instance is computed, validated, and recorded. There is no separate runtime, no bespoke implementation library, no compiled binary that "executes" the primitive — because the primitive is a semantic object, and semantic objects are executed by being well-formed-questioned to AI validators against the reference corpus.

5.2 Why This Is Not a Hand-Wave

A reasonable validator (and at least one validator in the PoW record did) reads "no executable specification" as a fatal flaw. The ERES position is precise: classical cryptographic primitives are executable as bit-level operations on byte-strings; ERES primitives are executable as well-formed-question operations on semantic-strings. The reduction is real, the operations are deterministic in the relevant sense (the same WFQ to the same AI corpus against the same ERES-GitHub state produces convergent — not bit-identical — outputs), and the verification method is observable: the MIEVM convergent rating is itself the test vector.

This is a category claim, not an evasion. The primitive is not failing to be a cryptographic primitive; it is succeeding at being a different kind of primitive — a **semantic primitive** whose executability runs on an AI substrate rather than a CPU substrate. The substrate disagreement is what produces the validator-frame disagreement seen in the PoW record (cryptography-frame vs. systems-doc-frame vs. layered-architecture-frame). The reduction stated here is the architectural commitment that resolves the disagreement: ERES does not claim to be classical cryptography; ERES claims to be its own primitive class, executable on an AI-validator substrate.

5.3 The MIEVM Worked Example

The convergent validation that produced the PoW record driving this very revision is itself the canonical worked example of the WFQ × AI + ERES-GitHub reduction:

1. **WFQ:** "rate independently and collectively" against the three slot deposits.
2. **AI:** DeepSeek, Grok, ChatGPT (three of the four MIEVM short-list members).
3. **ERES-GitHub:** the documents themselves were the corpus snapshot under evaluation.
4. **Output:** three independent ratings plus three collective ratings, varying in absolute score (4.2–9.2 range) but converging on the same three weakness categories (security model, implementation reduction, encoding standard).
5. **Convergence as test vector:** the agreement across validators on what is missing is the test vector that drove this v1.4 revision.

The reduction is not a future capability; it has already been operating, and v1.4 makes the operation explicit. Future ERES primitive instances are validated by the same mechanism: WFQ

posed to MIEVM-class validators against the ERES-GitHub corpus, with convergent output as the operational result.

5.4 Test Vector Generation Procedure

For a future revision (v1.5 or later) or for any downstream primitive instance, the canonical test vector generation procedure is:

1. Compose a WFQ targeting the specific primitive property to be tested.
2. Submit the WFQ to at least three MIEVM-class validators independently.
3. Record the convergent output along with disagreement spread.
4. Compute the convergent mean and the spread metric.
5. Record the result as a PoW deposit (e.g., MIEVM-PoW-2026-001-NNN) with: WFQ text, validator outputs, convergent mean, spread, GitHub corpus state hash.

Convergent mean ≥ 8.5 with spread ≤ 2.5 is the canonical "test passes" threshold per the MIEVM convergent validation rubric.

6. The Three-Tier Stack (Retained from v1.3)

The Symbols / Characters / Talonics three-tier stack is preserved without modification. Symbols are the atomic glyph layer; Characters are the compositional identity layer; Talonics are the governance-state layer (452 elements, 12 parts, $18 \times 7 = 126$ governance states). The stride encoding from §4 operates over Talonic state transitions.

7. The VEILED State (Retained from v1.3, Lifted to Stride Encoding)

VEILED replaces $\perp$ in the ERES Stack Primitive. v1.4 lifts the VEILED state into the stride encoding: VEILED corresponds to a 0 in the IDIPITIS binary stride. A VEILED attestation is a faithful attestation operating on a 0-stride; an absent attestation is a stride-incomplete attestation; a denied attestation is an explicit negation surface separate from the stride. The three are now distinguishable not only semantically (per v1.3) but structurally (per v1.4).

8. Trifurcated Math (Retained from v1.3)

Trifurcated Math remains the third pedagogical isomorph of Triune Math Key 2 ($M \times E + C = R$), exposing Matter, Energy, and Constant as separable but co-determinative inputs to Reason. The three Triune Math keys are unchanged: Key 1 ($C = R \times P / M$), Key 2 ($M \times E + C = R$), Key 3 ($REAL = (E \cdot M \cdot R) / (T \cdot S)$).

9. The Primitive Class (Retained, Augmented)

The resonance-keyed semantic superposition primitive class is defined as the family of cryptographic objects O such that:

1. O is constructed over a three-tier semantic stack (Symbols / Characters / Talonics).
2. O 's operational relation is of the form (council $\times$ compatibility) + audit = output, with output decomposed into seven attestation dimensions.
3. O admits a VEILED state distinct from rejection.
4. O 's pedagogical exposition is consistent with the three Triune Math isomorphs and Trifurcated Math.
5. **(v1.4 addition)** O preserves the IDIPITIS binary stride invariant under all faithful operations and exhibits stride collapse only under named adversary classes.
6. **(v1.4 addition)** O 's executable specification is the WFQ $\times$ AI + ERES-GitHub reduction, with test vectors generated by MIEVM convergent validation.

10. Position in the ERES Stack

The Primitive remains the upstream dependency for EAAP v1.3 (slot 432), SCALULAR, and Meritcoin Proof-of-Resonance. With v1.4, downstream protocols can cite a Security Model rather than infer one, and the implementation reduction propagates: EAAP attestations are themselves WFQ $\times$ AI + ERES-GitHub instances at the protocol layer.

11. Conclusion

v1.4 closes the gaps the MIEVM convergent rating identified without abandoning any of v1.3's load-bearing commitments. The IDIPITIS Security Model gives the primitive an explicit threat model, invariant pattern, and failure conditions. IDIPITIS names the corruption mode as palindromic inverse. The 111000111 / 000111000 stride encodes the invariant in nine bits. The WFQ $\times$ AI + ERES-GitHub reduction makes the primitive's executable specification explicit. CBGMODD and FAVORS are locked at canonical expansion. The primitive remains a semantic primitive on an AI-validator substrate — successfully so, by the criteria that primitive class earns rather than borrows.

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